Q1 - 2024 (04 Apr Shift 1)

Let $f : \mathbf{R} \to \mathbf{R}$ be a function given by

$$f(x)=\begin{cases}\frac{1-\cos 2x}{x^2}, & x<0\\ \alpha, & x=0,\\ \frac{\beta\sqrt{1-\cos x}}{x}, & x>0\end{cases}$$

where $\alpha, \beta \in \mathbf{R}$. If f is continuous at $x=0$, then $\alpha^2+\beta^2$ is equal to :

(1) 3

(2) 12

(3) 48

(4) 6

Q2 - 2024 (04 Apr Shift 2)

If the function

$$f(x)=\begin{cases}\frac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}}, & x\neq 0\\ a\log_e 2\log_e 3 & ,x=0\end{cases}$$

is continuous at $x=0$, then the value of a^2 is equal to

(1) 968

(2) 1152

(3) 746

(4) 1250

Q3 - 2024 (05 Apr Shift 1)

If the function $f(x)=\frac{\sin 3x+\alpha\sin x-\beta\cos 3x}{x^3}, x\in\mathbf{R}$, is continuous at $x=0$, then $f(0)$ is equal to :

(1) 2

(2) -2

(3) 4

(4) -4

Q4 - 2024 (05 Apr Shift 2)

Let $f:[-1,2] \rightarrow \mathbf{R}$ be given by $f(x)=2x^2+x+\left[x^2\right]-[x]$, where $[t]$ denotes the greatest integer less than or equal to t. The number of points, where f is not continuous, is :

(1) 5

(2) 6

(3) 3

(4) 4

Q5 - 2024 (06 Apr Shift 2)

Let $[\mathrm{t}]$ denote the greatest integer less than or equal to t. Let $f:[0, \infty) \rightarrow \mathbf{R}$ be a function defined by $f(x)=\left[\frac{x}{2}+3\right]-[\sqrt{x}]$. Let S be the set of all points in the interval $[0,8]$ at which f is not continuous. Then $\sum_{\mathrm{a} \in \mathrm{S}} \mathrm{a}$ is equal to ______

Q6 - 2024 (08 Apr Shift 2)

For $\mathrm{a}, \mathrm{b}>0$, let $f(x)=\begin{cases}\frac{\tan ((\mathrm{a}+1) x)+\mathrm{b} \tan x}{x}, & x<0 \\ 3, & x=0 \\ \frac{\sqrt{\mathrm{a} x+\mathrm{b}^2 x^2}-\sqrt{\mathrm{a} x}}{\mathrm{b} \sqrt{\mathrm{a}} x \sqrt{x}}, & x>0\end{cases}$

be a continous function at $x=0$. Then $\frac{\mathrm{b}}{\mathrm{a}}$ is equal to :

(1) 6

(2) 4

(3) 5

(4) 8

Q7 - 2024 (09 Apr Shift 1)

Let $f:(0, \pi) \rightarrow \mathbf{R}$ be a function given by $f(x)=$
$$\begin{cases}\left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2} \\ \mathrm{a}-8, & x=\frac{\pi}{2} \\ (1+|\cot x|)^{\frac{\mathrm{b}}{|\tan x|}}, & \frac{\pi}{2}<x<\pi\end{cases}$$
where $\mathrm{a}, \mathrm{b} \in \mathbf{Z}$. If f is continuous at $x=\frac{\pi}{2}$, then $\mathrm{a}^2+\mathrm{b}^2$ is equal to

Answer Key

Q1 (2)	**Q2** (2)	**Q3** (4)	**Q4** (4)
Q5 (17)	**Q6** (1)	**Q7** (81)	

Q1

$$f\left(0^{-}\right)=\lim_{x\rightarrow0^{-}}\frac{2\sin^{2}x}{x^{2}}=2=\alpha$$

$$f\left(0^{+}\right)=\lim_{x\rightarrow0^{+}}\beta\times\sqrt{2}\frac{\sin\frac{x}{2}}{2\frac{x}{2}}=\frac{\beta}{\sqrt{2}}=2$$

$\Rightarrow\beta=2\sqrt{2}$
$\alpha^{2}+\beta^{2}=4+8=12$

Q2

$$\lim_{x\rightarrow0}f(x)=a\ln2\ln3$$

$$\lim_{n\rightarrow0}\frac{72^{x}-9^{x}-8^{x}+1}{\sqrt{2}-\sqrt{1+\cos x}}=\lim_{x\rightarrow0}\frac{\left(8^{x}-1\right)\left(9^{x}-1\right)}{\sqrt{2}-\sqrt{1+\cos x}}$$

$$\lim_{n\rightarrow0}\left(\frac{8^{x}-1}{x}\right)\left(\frac{9^{x}-1}{x}\right)\left(\frac{x^{2}}{1-\cos x}\right)(\sqrt{2}+\sqrt{1+\cos x})$$

$\therefore\ln8\times\ln9\times2\times2\sqrt{2}=24\sqrt{2}\ln2\ln3$
$\therefore a=24\sqrt{2},a^{2}=576\times2=1152$

Q3

$$f(x)=\frac{\sin3x+\alpha\sin x-\beta\cos3x}{x^{3}}$$

is continuous at x = 0

$$\lim_{x\rightarrow0}=\frac{3x-\frac{(3x)^{3}}{\underline{|3}}+\ldots+\alpha\left(x-\frac{x^{3}}{\underline{|3}}\ldots\right)-\beta\left(1-\frac{(3x)^{2}}{\underline{|2}}\ldots\right)}{x^{3}}=f(0)$$

$$\lim_{x\rightarrow0}=\frac{-\beta+x(3+\alpha)+\frac{9\beta x^{2}}{\underline{|2}}+\left(\frac{-27}{\underline{|3}}-\frac{\alpha}{\underline{|3}}\right)x^{3}\ldots}{x^{3}}=f(0)$$

for exist

$$\beta=0,3+\alpha=0,-\frac{27}{\underline{|3}}-\frac{\alpha}{\underline{|3}}=f(0)$$

$$\alpha=-3,-\frac{27}{6}-\frac{(-3)}{6}=f(0)$$

$$f(0)=\frac{-27+3}{6}=-4$$

Q4

Doubtful points : $-1, 0, 1, \sqrt{2}, \sqrt{3}, 2$ at $x = \sqrt{2}, \sqrt{3}$

$$f(x) = \underbrace{\left(2x^2 + x - [x]\right)}_{\text{Cont.}} + \underbrace{\left[x^2\right]}_{\text{Cont.}} = \text{Discount}$$

at $x = -1$:

$$\left.\begin{array}{ll}\text{RHL} \Rightarrow & f(x) = (2 - 1 - (-1)) + 0 = 2 \\ & f(-1) = 2 - 1 - (-1) + 1 = 3\end{array}\right\} \text{Dis.}$$

at $x = 2$:

$$\left.\begin{array}{lll}\text{LHL} \Rightarrow f(x) & = 8 + 2 - 1 + 3 = 12 \\ \quad\quad\quad f(2) & = 8 + 2 - 2 + 4 = 12\end{array}\right\} \text{Cont.}$$

at $\mathrm{x} = 0$:

$$\left.\begin{array}{ll}\text{LHL} \Rightarrow & 0 + 0 - (-1) + 0 = 1 \\ & \mathrm{f}(0) = 0\end{array}\right\} \text{Dis.}$$

at $\mathrm{x} = 1$

$$\left.\begin{array}{ll}\text{LHL} \Rightarrow & 2 + 1 - 0 + 0 = 3 \\ & f(1) = 3 - 1 + 1 = 3 \\ \text{RHL} \Rightarrow & 2 + 1 - 1 + 1 = 3\end{array}\right\} \text{Cont.}$$

Q5

$\left[\frac{x}{2} + 3\right]$ is discontinuous at $x = 2, 4, 6, 8$

$\sqrt{\mathrm{x}}$ is discontinuous at $\mathrm{x} = 1, 4$ $\mathrm{F}(\mathrm{x})$ is discontinuous at $\mathrm{x} = 1, 2, 6, 8$

$\sum \mathrm{a} = 1 + 2 + 6 + 8 = 17$

Q6

$$\lim_{x \to 0} f(x) = f(0) = 3$$

$$\lim_{x \to 0^+} \frac{\sqrt{ax + b^2x^2} - \sqrt{ax}}{b\sqrt{a}x\sqrt{x}} = 3$$

$$\lim_{x \to 0^+} \frac{ax + b^2x^2 - ax}{b\sqrt{a}x^{3/2}\left(\sqrt{ax + b^2x^2} + \sqrt{ax}\right)}$$

$$\lim_{x \to 0^+} \frac{b^2}{b\sqrt{a}\left(\sqrt{a + b^2x} + \sqrt{a}\right)}$$

$$\frac{b}{\sqrt{a} \cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \Rightarrow \frac{b}{a} = 6$$

Q7

LHL at x $= \frac{\pi}{2}$

$\lim_{x \to \frac{\pi}{2}} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}} = \left(\frac{8}{7}\right)^{0} = 1$

RHL at x $= \frac{\pi}{2}$

$\lim\limits_{x \to \frac{\pi}{2}} (1 + |\cot x|)^{\frac{b}{\operatorname{lan}} \tan x}$

$= e^{\lim_{x - \frac{\pi}{2}} |\cot x| \frac{b}{a} |\tan x|} = e^{\frac{b}{a}}$

$\Rightarrow 1 = a - 8 = e^{\frac{b}{a}}$

$\Rightarrow a = 9, b = 0$

$\Rightarrow a^2 + b^2 = 81$